7

Numerical Integration

Chapter Learning Outcomes

CLO

7

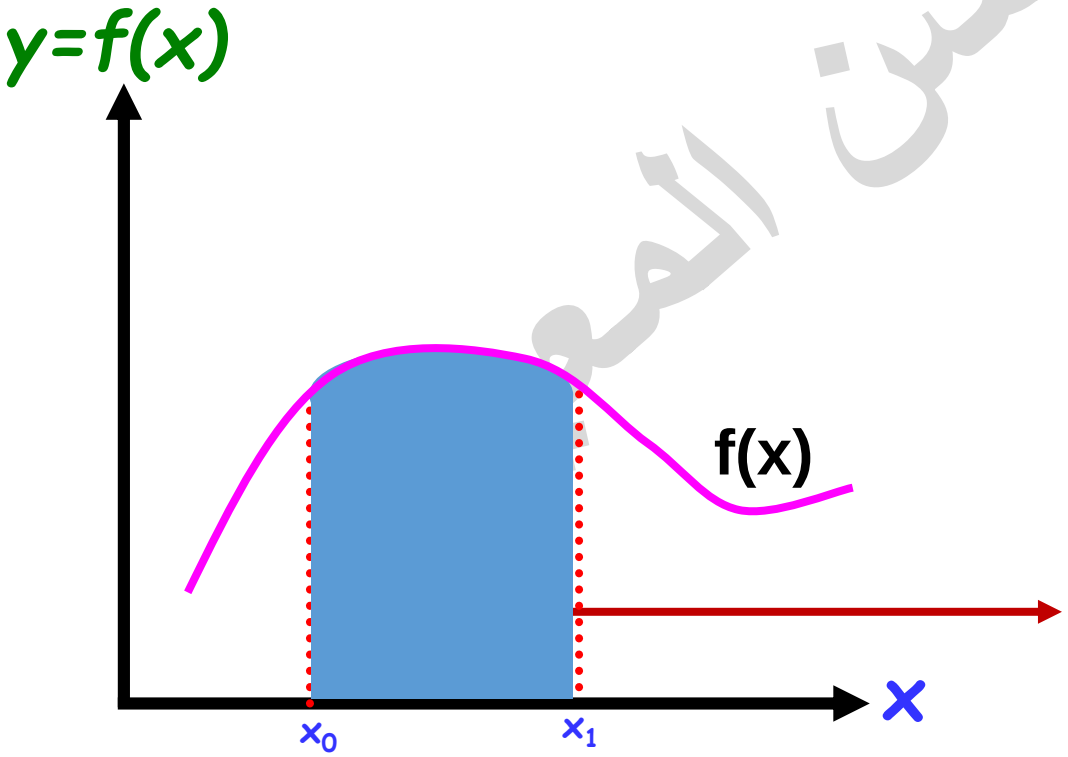
- Know and understand the meaning of Integration.
- Use (Trapezoidal Rule, Simpson's 1/3 and Simpson's 3/8) rules to approximate the integral of a function based on a given data.

حسن المعيدي

What is Integration?

Integration is the process of measuring the area under a function plotted on a graph.

Integral = area under the function

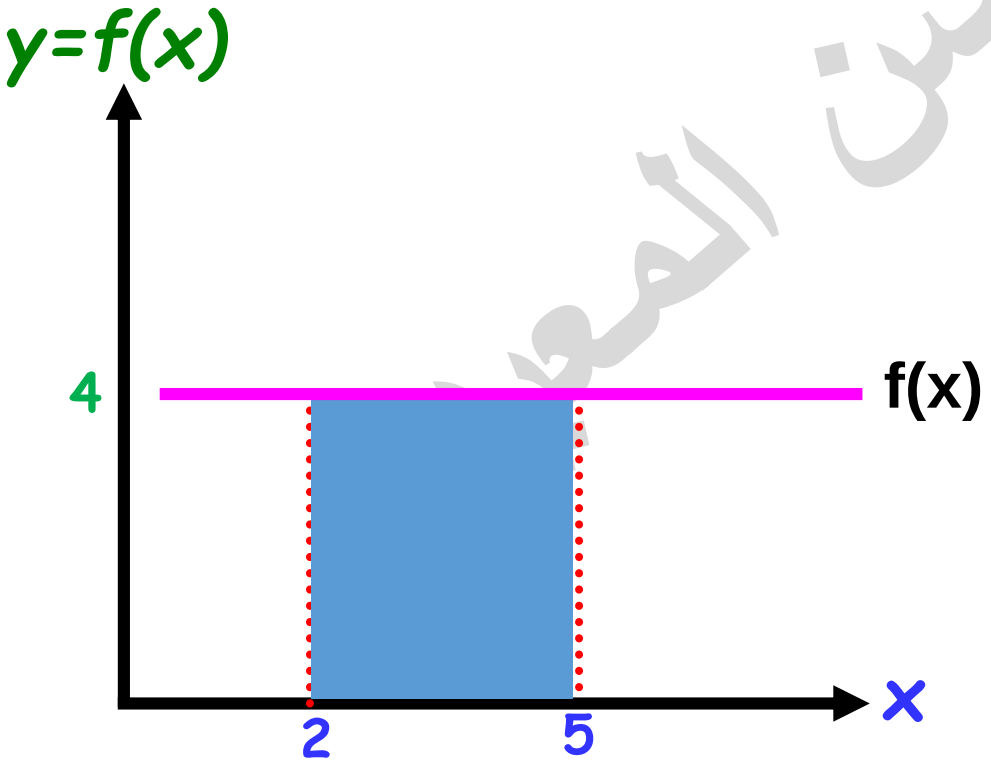


$$Area = \int_{x_0}^{x_1} f(x).dx$$

What is Integration?

Example: If f(x)=4, find $\int_2^5 f(x).\,dx$

Integral = area under the function



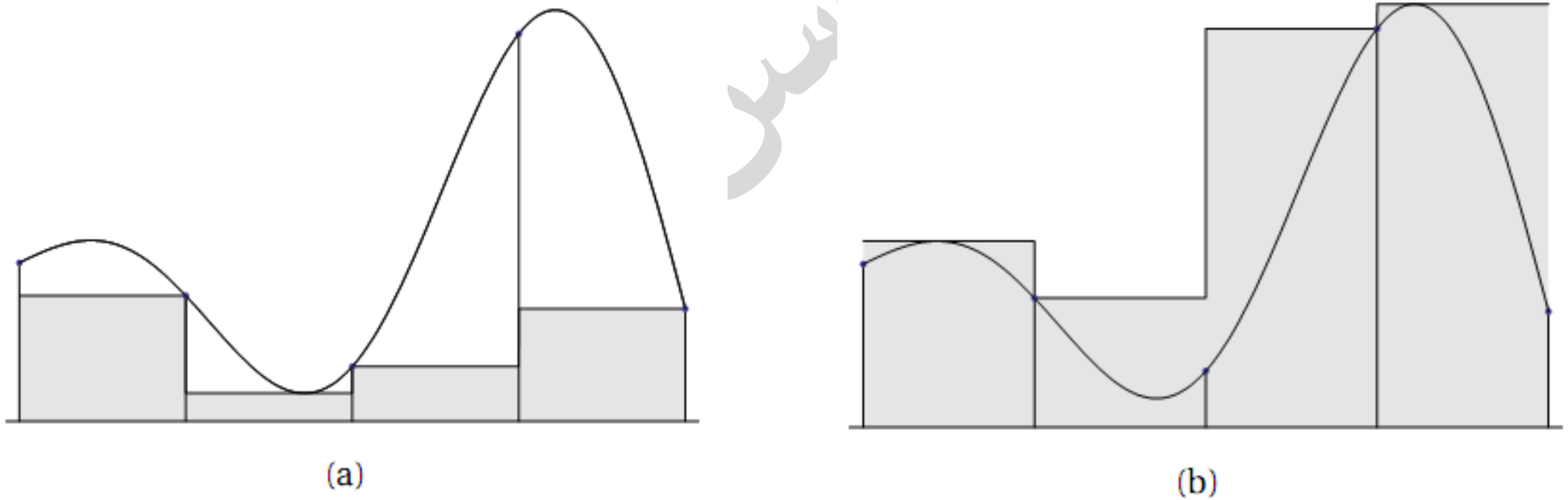
1 – Area for Rectangle

2 – Integration Rule

$$\int_2^5 4.\,dx$$

What is Integration?

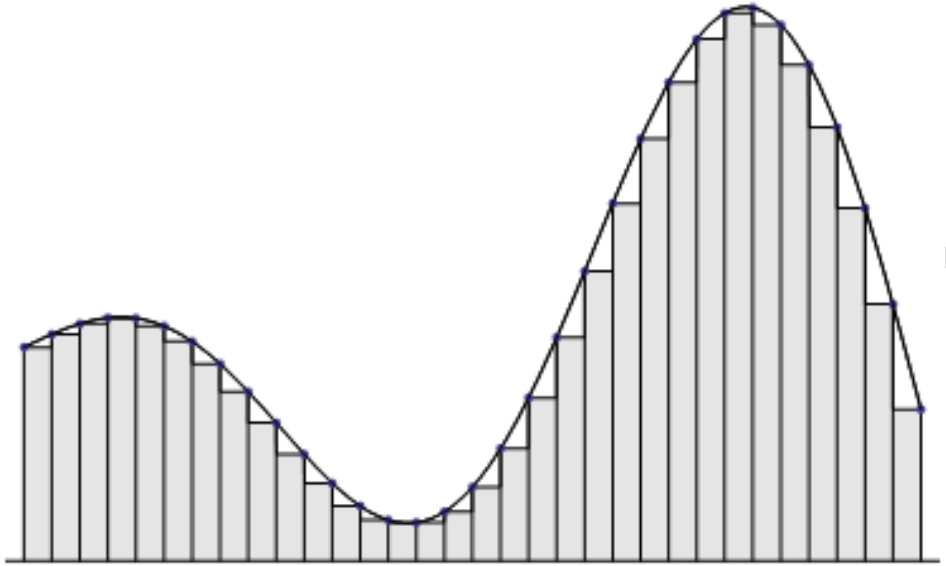
Integral = area under the function



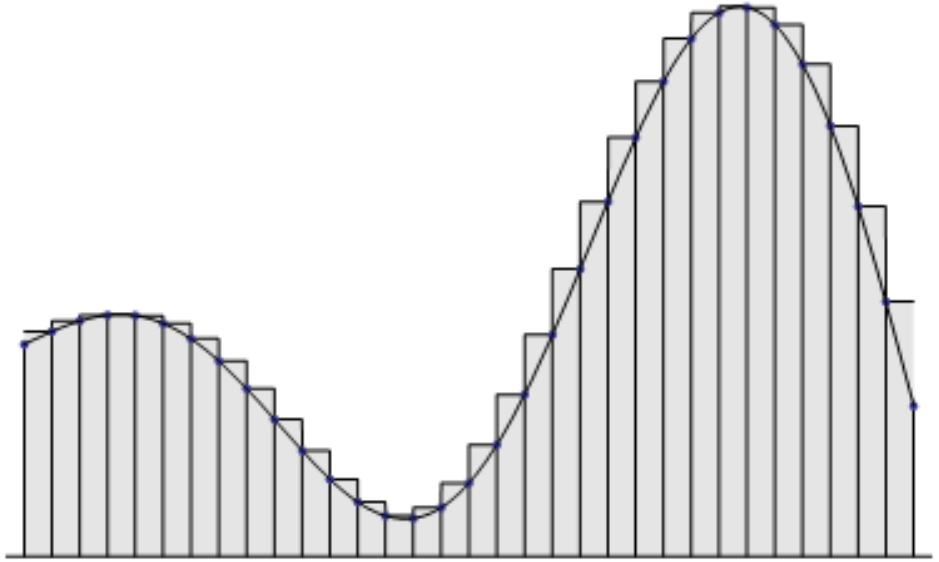
How to minimize the error?

What is Integration?

We can minimize the error by increasing the number of subinterval



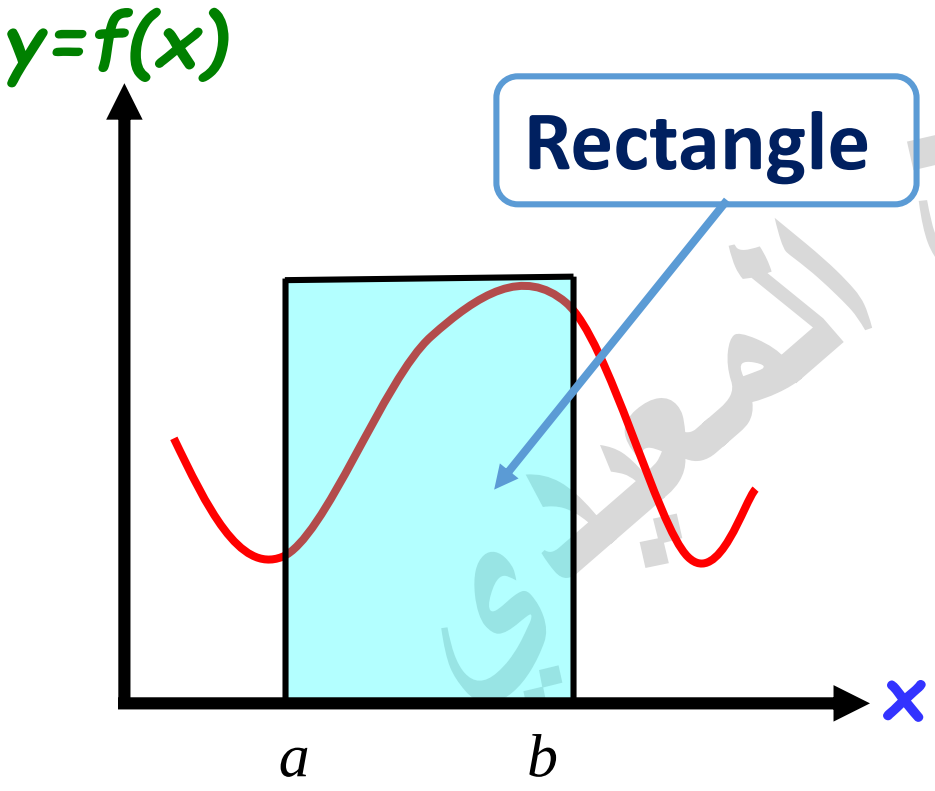
(c)

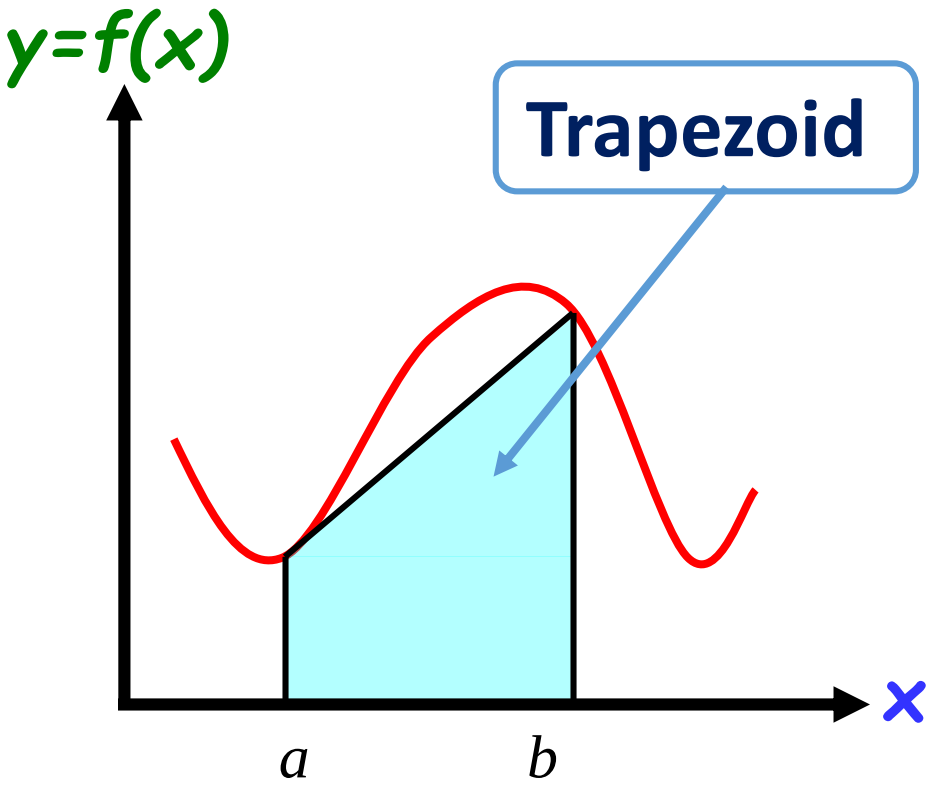


(d)

What is Integration?

Which one is the best?





د. حسن المعيدي

Numerical Integration Rules

Trapezoidal Rule	n = 1
Composite Trapezoidal Rule	n = 2, 3, 4, 5, …
Simpson's (1/3) Rule	n = 2
Composite Simpson's (1/3) Rule	n = 4, 6, 8, 10, …
Simpson's (3/8) Rule	n = 3
Composite Simpson's (3/8) Rule	n = 6, 9, 12, 15, …

Numerical Integration Ruls

Trapezoidal Rule	n = 1

$$\int_{x_0}^{x_1} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_1)\right]$$

Composite Trapezoidal Rule	n = 2, 3, 4, 5, …

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_n) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \ldots\right]$$

Numerical Integration Ruls

Simpson's (1/3) Rule

n = 2

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}\left[f(x_0) + f(x_2) + 4f(x_1)\right]$$

Composite Simpson's (1/3) Rule

n = 4, 6, 8, 10, …

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{3}\left[f(x_0) + f(x_n) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \ldots\right]$$

Numerical Integration Ruls

Simpson's (3/8) Method

n = 3

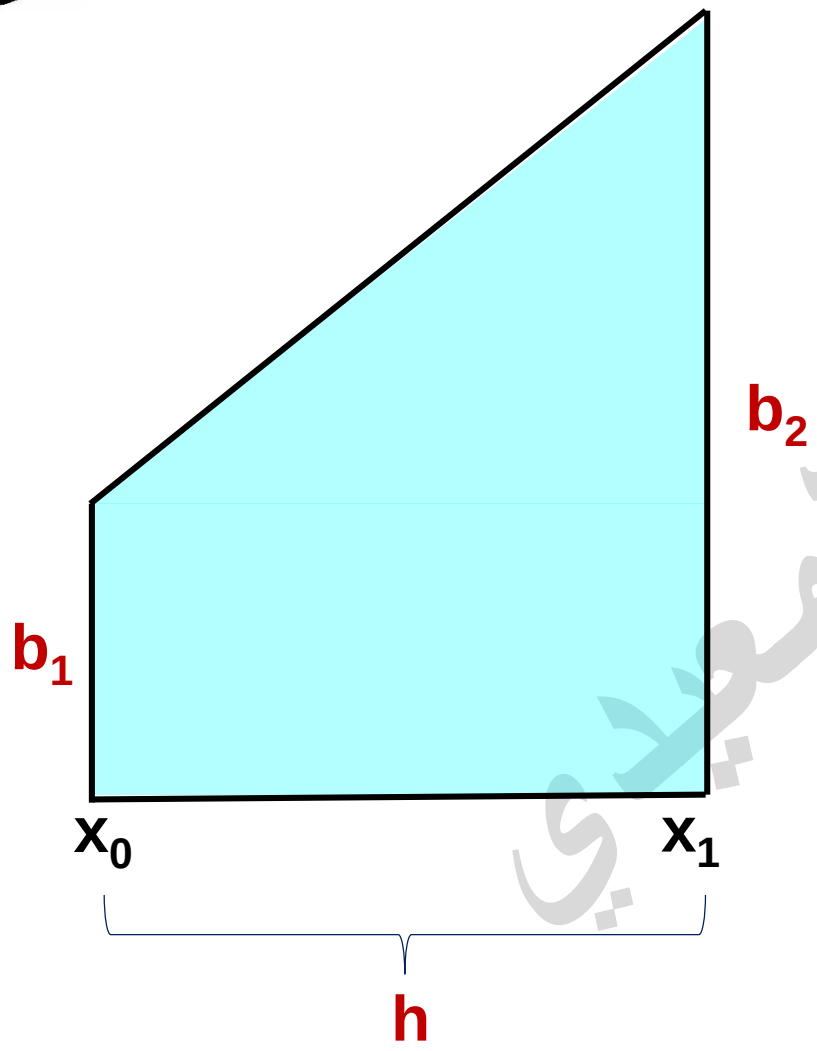
$$\int_{x_0}^{x_3} f(x)\,dx = \frac{3h}{8}\Big[f(x_0) + f(x_3) + 3(f(x_1) + f(x_2))\Big]$$

Composite Simpson's (3/8) Method

n = 6, 9, 12, 15, …

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{3h}{8}\Big[f(x_0) + f(x_n) + 3(f_1 + f_2 + f_4 + \ldots) + 2(f_3 + f_6 + f_9 + \ldots)\Big]$$

Area of a Trapezoid



Area for Trapezoid

$$= \frac{1}{2} \times (b_1 + b_2) \times h$$

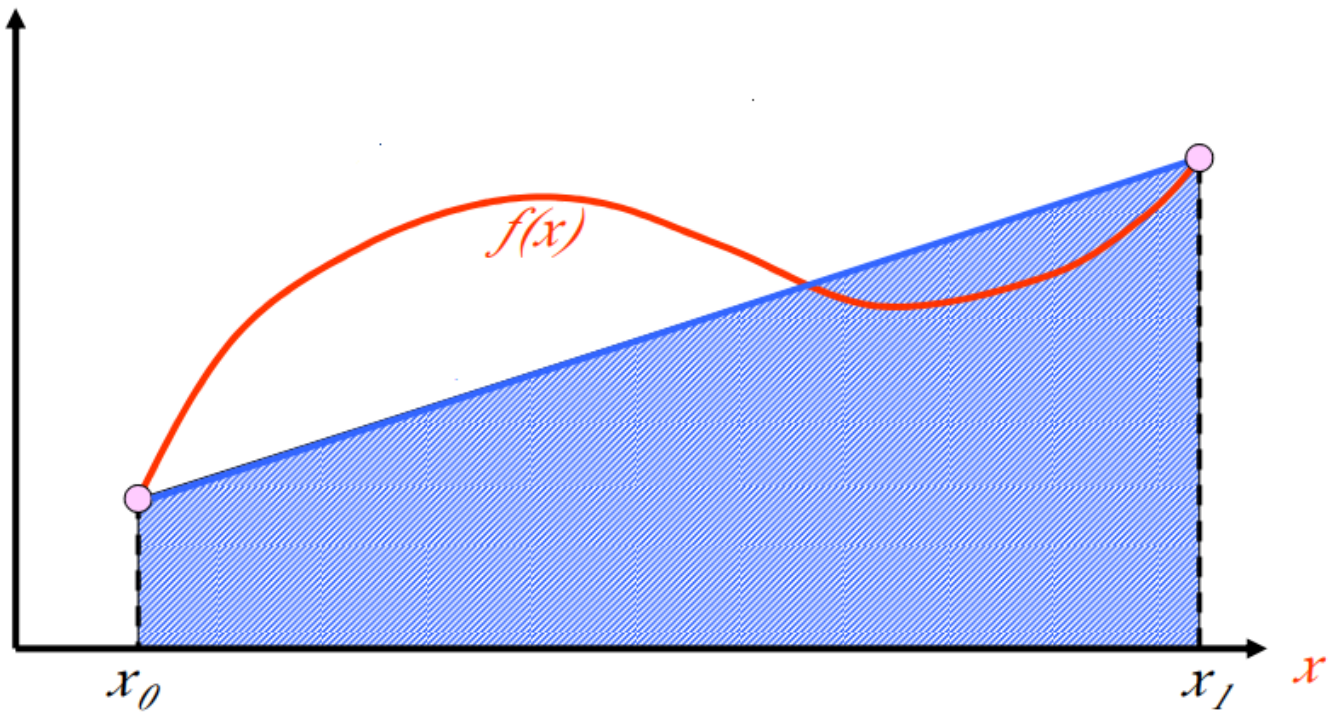
$$= \frac{h}{2}(b_1 + b_2)$$

د. حسن المعيدي

Trapezoidal Rule

No. of Segments (n) = 1

$$h = \frac{x_1 - x_0}{1}$$



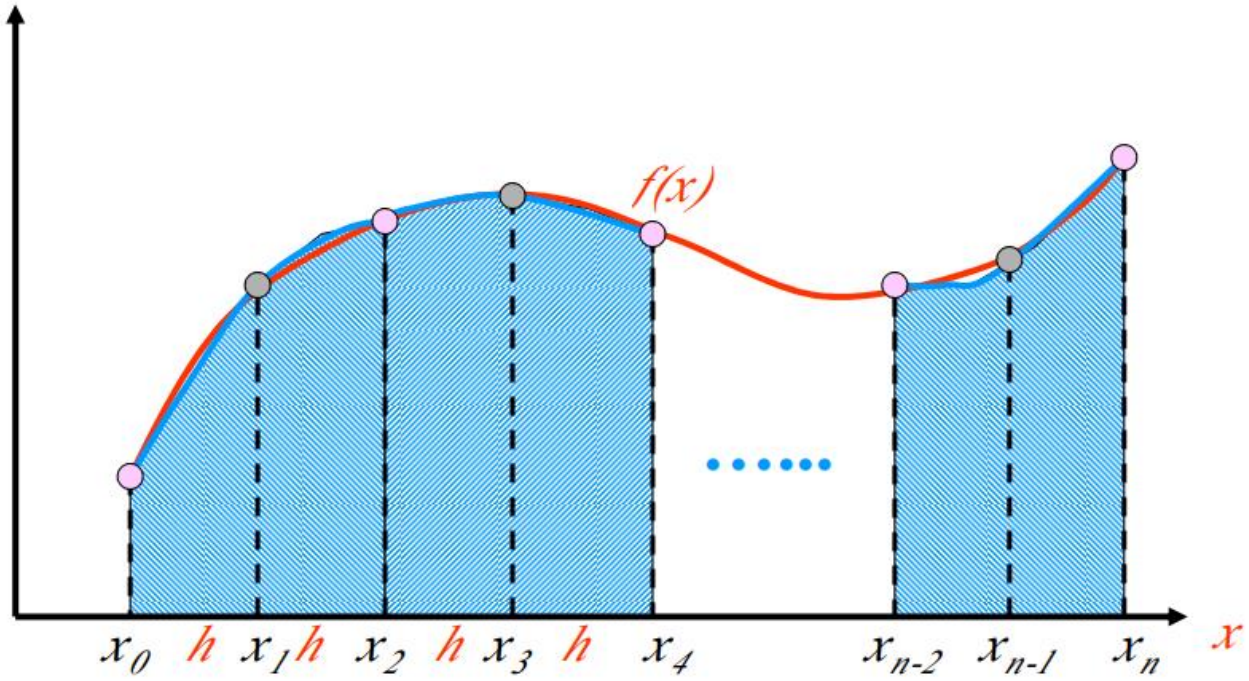
$$\int_{x_0}^{x_1} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_1)\right]$$

المعيدي

Composite Trapezoidal Rule

No. of Segments (n) = n

$$h = \frac{x_n - x_0}{n}$$



$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{2}\left[f(x_0) + f(x_n) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + \ldots\right]$$

Example 1

Approximate $\int_{1}^{3} x^3 \, dx$ using Trapezoidal rule with 1 sub-interval.

د. حسن المعيدي

Example 2

Approximate $\int_{1}^{3} x^3 \, dx$ using Trapezoidal rule with 2 sub-intervals.

د. حسن المعيدي

Example 3

Approximate $\int_0^4 (1 - e^{-2x})\, dx$ using Trapezoidal rule with 4 segments.

د. حسن المعيدي

Example 4

Given the values for x and $f(x)$ in the following table, use the Trapezoidal rule to approximate $\int_{2.1}^{3.6} f(x)\,dx$.

	x_0	x_1	x_2	x_3	x_4	x_5
x	2.1	2.4	2.7	3.0	3.3	3.6
$f(x)$	3.2	2.7	2.9	3.5	4.1	5.2

المعيدي

Example 5

Given the values for x and $f(x)$ in the following table, use the composite trapezoidal rule (with 5 segments) to approximate $\int_{2.1}^{3.6} f(x)\,dx$.

	x_0	x_1	x_2	x_3	x_4	x_5
x	2.1	2.4	2.7	3.0	3.3	3.6
$f(x)$	3.2	2.7	2.9	3.5	4.1	5.2

د. حسن المعيدي

Numerical Integration Methods

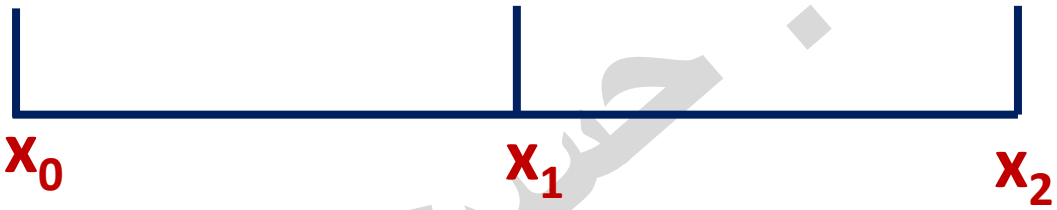
- Trapezoidal Method ✓
- Composite Trapezoidal Method ✓

- Simpson's (1/3) Method
- Composite Simpson's (1/3) Method

- Simpson's (3/8) Method
- Composite Simpson's (3/8) Method

Simpson's (1/3) Rule

No. of Segments (n) = 2



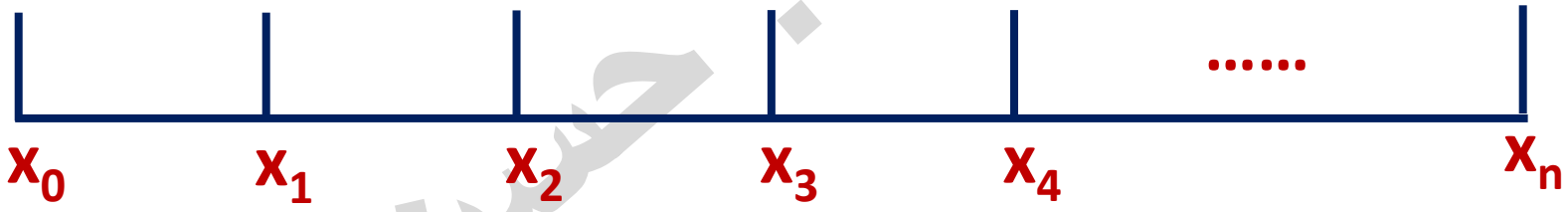
$$h = \frac{x_2 - x_0}{2}$$

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}\left[f(x_0) + f(x_2) + 4f(x_1)\right]$$

Composite Simpson's (1/3) Rule

No. of Segments (n) = 4, 6, 8, ….

n is an even ≥ 4

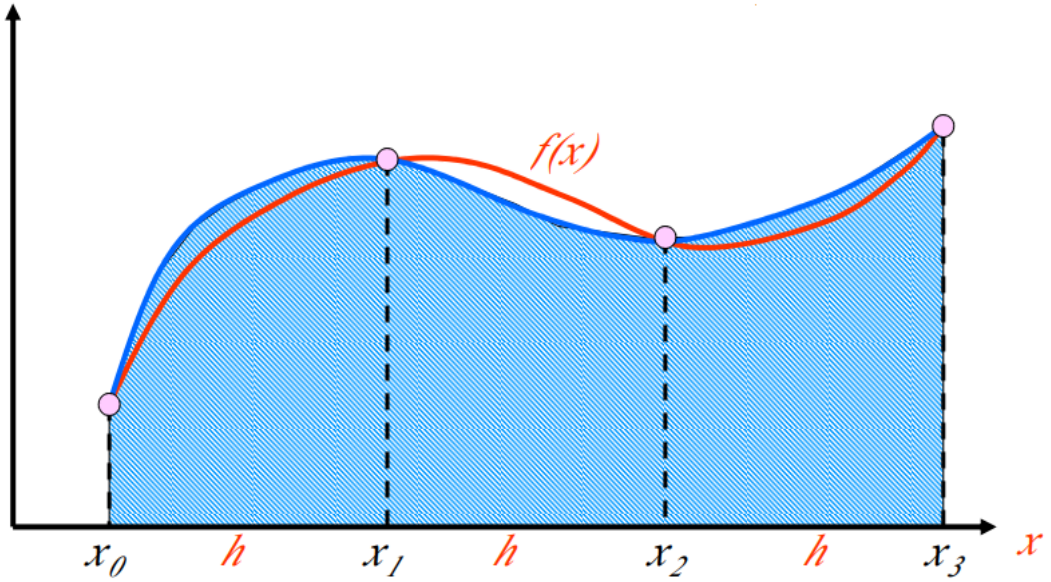


$$h = \frac{x_n - x_0}{n}$$

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{3}\left[f(x_0) + f(x_n) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \ldots\right]$$

Simpson's (3/8) Rule

No. of Segments (n) = 3



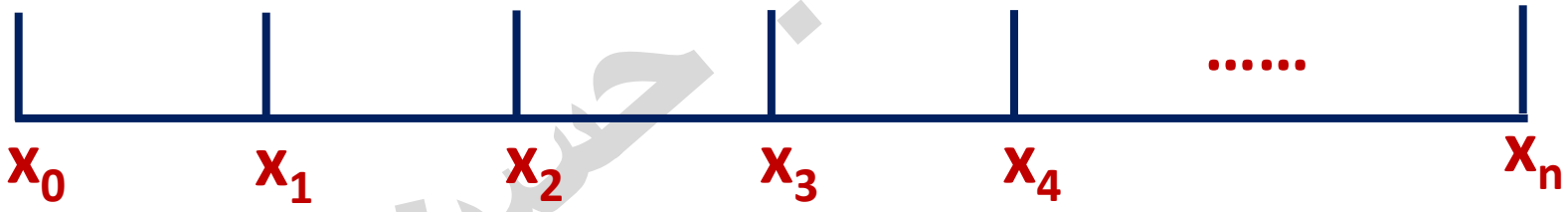
$$h = \frac{x_3 - x_0}{3}$$

$$\int_{x_0}^{x_3} f(x)\,dx = \frac{3h}{8}\left[f(x_0) + f(x_3) + 3(f(x_1) + f(x_2))\right]$$

Composite Simpson's (3/8) Rule

No. of Segments (n) = 6, 9, 12, ….

n is multiple of 3 ≥ 6



$$h = \frac{x_n - x_0}{n}$$

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{3h}{8}\left[f(x_0) + f(x_n) + 3(f_1 + f_2 + f_4 + \ldots) + 2(f_3 + f_6 + f_9 + \ldots)\right]$$

Example 6

Given the values for x and $f(x)$ in the following table, use the Simposon's (1/3) rule to approximate $\int_{2.1}^{3.3} f(x)\,dx$.

	x_0	x_1	x_2	x_3	x_4
x	2.1	2.4	2.7	3.0	3.3
$f(x)$	3.2	2.7	2.9	3.5	4.1

المعيدي

Example 7

Given the values for x and $f(x)$ in the following table, use the composite Simpson's (1/3) rule (with 4 segments) to approximate $\int_{2.1}^{3.3} f(x)\,dx$.

	x_0	x_1	x_2	x_3	x_4
x	2.1	2.4	2.7	3.0	3.3
$f(x)$	3.2	2.7	2.9	3.5	4.1

المعيدي

Example 8

Integrate the data using Simpson's–1/3 and Simpson's–3/8 Rules.

i	0	1	2	3	4	5	6	7
x_i	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7
f_i	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828

حسن المعيدي

Example 8

Integrate the data using Simpson's–1/3 and Simpson's–3/8 Rules.

i	0	1	2	3	4	5	6	7
x_i	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7
f_i	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828

حسن المعيدي

Example 8

Integrate the data using Simpson's–1/3 and Simpson's–3/8 Rules.

i	0	1	2	3	4	5	6	7
x_i	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7
f_i	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828

حسن المعيدي

Example 8

Integrate the data using Simpson's–1/3 and Simpson's–3/8 Rules.

i	0	1	2	3	4	5	6	7
x_i	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7
f_i	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828

حسن المعيدي

Example 9

Compute the value of

$$\int_{0.2}^{1.4}(\sin x - \log x + e^{x})\,dx, \textit{using composite Simposon's } \frac{3}{8} \textit{rule}.$$

د. حسن المعيدي

Example 9

د. حسن المعيدي

Check - Chapter Learning Outcomes

CLO

7

- Know and understand the meaning of Integration.
- Use (Trapezoidal Rule, Simpson's 1/3 and Simpson's 3/8) rules to approximate the integral of a function based on a given data.

حسن المعيدي

End of Chapter

7

نهاية المادة

أتمنى لكم كل التوفيق